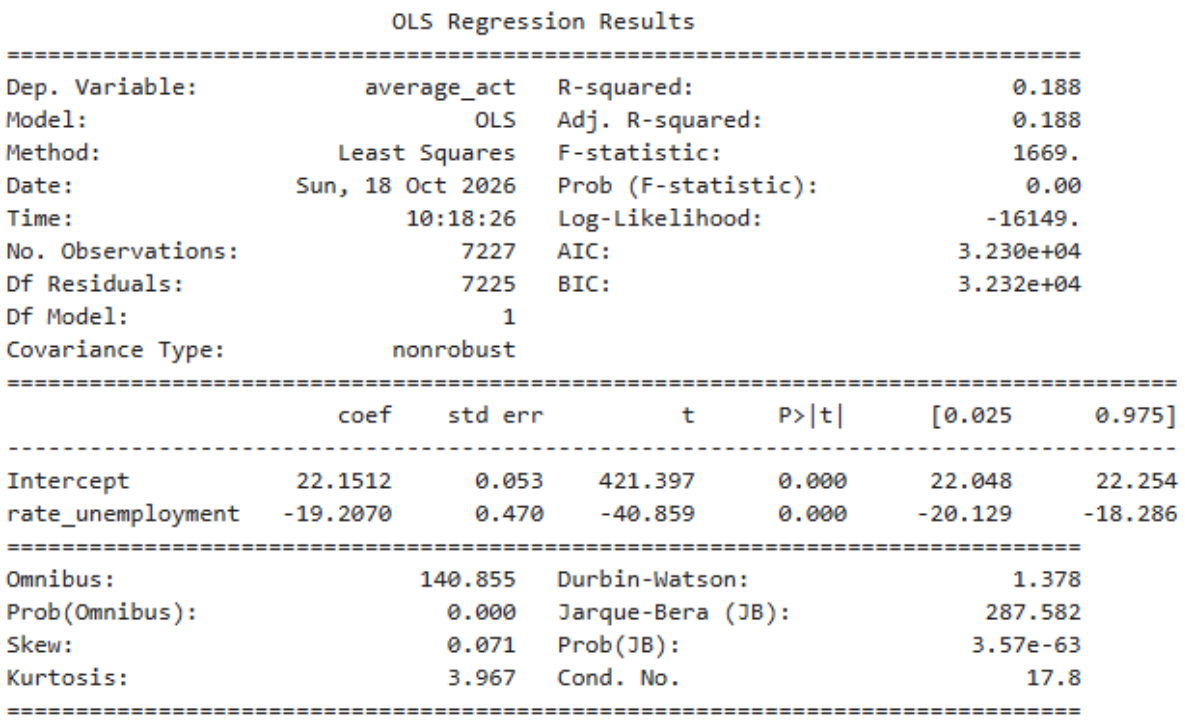
For the Education project, I used two datasets to answer the question: Does the community you live in affect the ACT scores that you may get? The datasets that were used for my project were EdGap and CCD. The EdGap dataset gave the five socioeconomic factors that are used in this study to compare with the other dataset that I used, CCD. This dataset gave information about the schools and students and what their ACT scores were. For this project, these two datasets had to be combined together to make a larger dataset so that it would be easier to do the regression. But before that, we had to clean up the data and get rid of any columns that were not important for the project.

While thinking about the question for this project and needing to use one of the five socioeconomic factors, I wanted to know the comparison between people who grew up in lower-income places and places with higher unemployment, and whether their ACT scores would be affected by these circumstances. Once the data was clean and I could do a regression, I chose to use the examples I saw in class and the results I got from my before-class assignment and put them together. Since in class we did financial income and how that affects ACT scores, we also saw that income was a very big part of the results. When doing the reduced model, it was almost the same as if we had just done the normal, whole model.



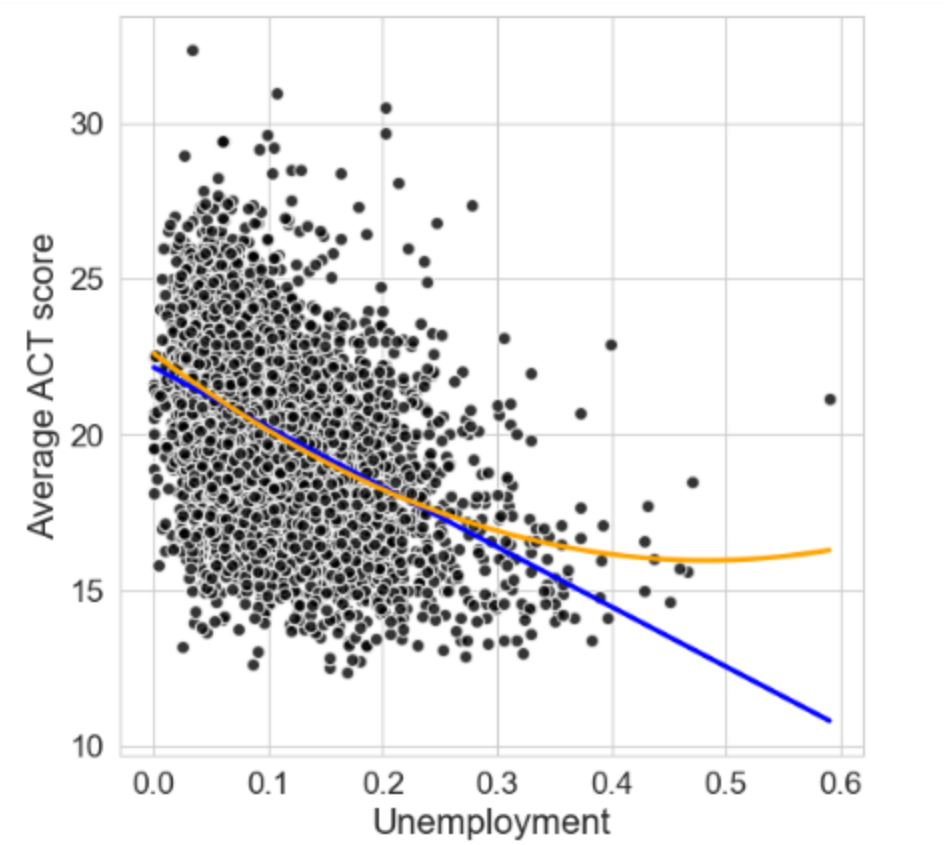
OLS Regression Results

Dep. Variable:	average_act	R-squared:	0.188
Model:	OLS	Adj. R-squared:	0.188
Method:	Least Squares	F-statistic:	1669.
Date:	Sun, 18 Oct 2026	Prob (F-statistic):	0.00
Time:	10:18:26	Log-Likelihood:	-16149.
No. Observations:	7227	AIC:	3.230e+04
Df Residuals:	7225	BIC:	3.232e+04
Df Model:	1		
Covariance Type:	nonrobust		

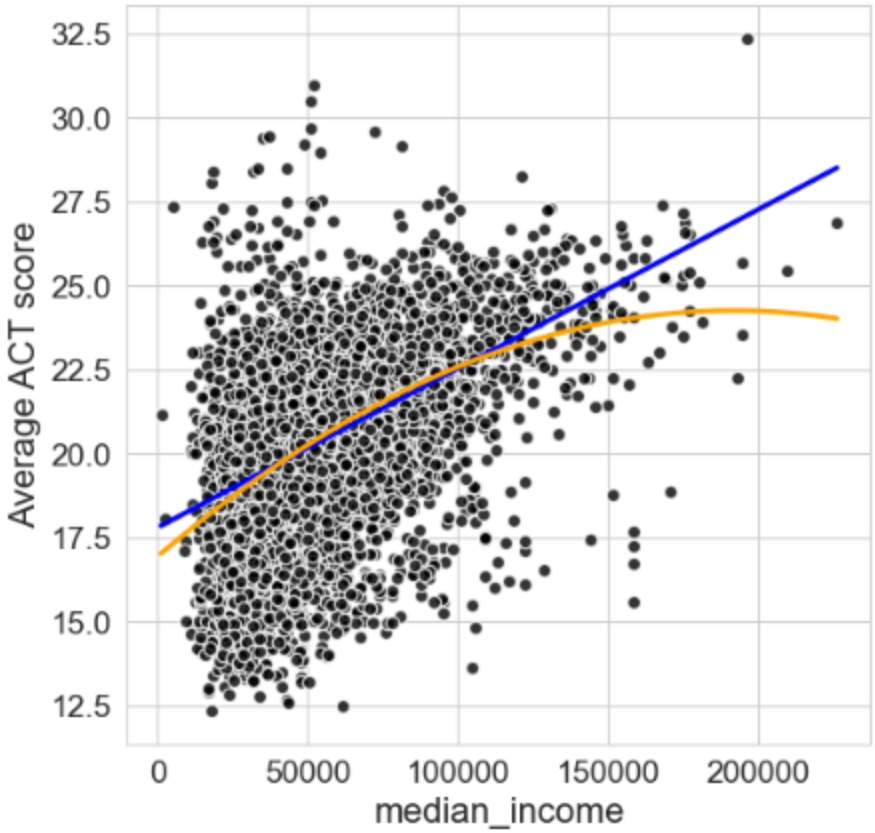
	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	22.1512	0.053	421.397	0.000	22.048	22.254
rate_unemployment	-19.2070	0.470	-40.859	0.000	-20.129	-18.286

Omnibus:	140.855	Durbin-Watson:	1.378
Prob(Omnibus):	0.000	Jarque-Bera (JB):	287.582
Skew:	0.071	Prob(JB):	3.57e-63
Kurtosis:	3.967	Cond. No.	17.8

The above photo is of the regression results for unemployment. The p-value is very small, and so is the R-squared value.



The scatter plot above shows that as unemployment gets higher, the ACT test scores seem to get lower as well.



The scatter plot above shows the effect of median income on ACT scores, and we can see that the more money there is, the higher the test scores usually are.

OLS Regression Results

Dep. Variable:	average_act	R-squared:	0.271
Model:	OLS	Adj. R-squared:	0.270
Method:	Least Squares	F-statistic:	1340.
Date:	Wed, 21 Oct 2026	Prob (F-statistic):	0.00
Time:	20:42:59	Log-Likelihood:	-15760.
No. Observations:	7227	AIC:	3.153e+04
Df Residuals:	7224	BIC:	3.155e+04
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	19.7055	0.099	199.442	0.000	19.512	19.899
rate_unemployment	-12.2934	0.507	-24.268	0.000	-13.286	-11.300
median_income	3.373e-05	1.18e-06	28.663	0.000	3.14e-05	3.6e-05

Omnibus:	233.186	Durbin-Watson:	1.369
Prob(Omnibus):	0.000	Jarque-Bera (JB):	640.056
Skew:	-0.033	Prob(JB):	1.03e-139
Kurtosis:	4.456	Cond. No.	1.18e+06

Notes:

The above photo is of the multiple regression of median income and unemployment. Once more, the p-value is small, and the R-squared is also very small if we compare it to the full regression of 0.628.

For this project, I used multiple regression charts and plots to find that it cannot be claimed that the community, using both income and unemployment, could truly affect the outcome of ACT scores. Median income had some effect, but nothing too drastic. Adding unemployment rates evened out the increase associated with median income, showing that, in the end, it doesn't really matter. The results are inconclusive together but apart it does affect the ACT score with unemployment, or with the median income.